biomethane by absorbing or scrubbing contaminants and pressurised for storage in high-pressure cylinders, gets upgraded to bio-CNG. Without proper purification, using biogas can lead to erosion of metal parts in vehicles. Purity of bio-CNG reserves in automobiles can be tracked using software and data analysis tools.

In India, there is immense scope for bio-CNG due to abundance of biomass. Built in 2016, the bio-CNG plant at Mahindra World City in Chennai converts hundred per cent of food and kitchen waste generated daily in the city into raw biogas. Bio-CNG thereby produced is utilised for cooking purposes, as a fuel for buses and tractors, as well as to power streetlights in the area.

In 2017, Tata Motors showcased the nation's first bio-CNG bus at the bio-energy programme, Urja Utsav.

Government policies are further boosting these initiatives. In October 2018, Sustainable Alternative Towards Affordable Transportation (SATAT) scheme was launched by Ministry of Petroleum and Natural Gas under which production of fifteen million tonnes of CBG from 5000 plants is expected by 2023.

Besides this, Galvanising Organic Bio-Agro Resources Dhan (GOBAR-DHAN) scheme, aimed at converting solid waste and cattle dung into compost, fertiliser, biogas and bio-CNG, was announced in Union Budget 2018-19.

Global adoption of bio-CNG depends on several economic, environmental and technical factors. Capital cost for installation of bio-CNG plants is high. Sources for feed materials are not consistent, and refuelling stations are more complicated than conventional ones since high pressure for various components needs to be maintained.

Due to its short driving range, additional fuel cylinders may be required for extending the range. There is a shortage of skilled professionals necessary for the production process in this field. Also, there is a lack of proper standards and regulations in many countries when it comes to installation, operation and maintenance of plants.

However, the plethora of advantages will drive the growth of bio-CNG vehicles, primarily trucks and buses. In Sweden, Bio-CNG is already being used on a large scale to run buses. UK-based John Lewis plans to switch all its heavy delivery trucks to biomethane-powered versions by 2028. Petrol as well as diesel engine driven vehicles can be driven by bio-CNG through retrofitting.

Bio-CNG can help with the problem of solid waste that is dumped in landfills and polluted air due to stubble-burning while simultaneously creating new employment opportunities.

There is a need to create awareness among masses about the benefits of biogas to maximise its use for various purposes including transportation and achieve sustainable development.

—Ayushee Sharma

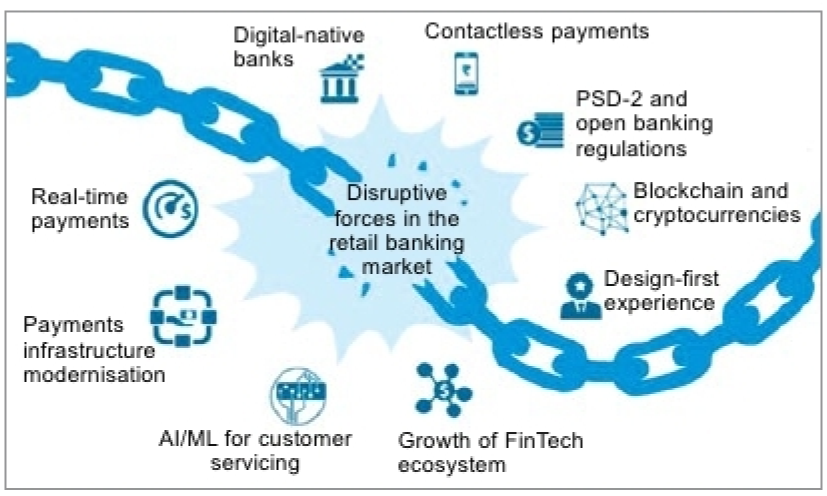
3 BECOMING FINANCIALLY SMARTER WITH NEW-AGE BANKING

While technology and simple regulations are blurring business boundaries and boosting employment, security concerns related to the use of algorithms need to be addressed through comprehensive legal frameworks

India's economic slowdown has been hitting the headlines recently. According to International Monetary Fund (IMF), economic growth was estimated to have declined globally to 3.3 per cent in 2019. Appropriate reforms, if not made, can lead to further decrease this year.

Banking has a vast impact on GDP of a nation. As small and medium businesses (SMBs) have insufficient collateral, their chances of not being able to repay on time are higher. This makes traditional institutions reluctant to lend them money to avoid financial risks.

In countries like China, economists have concluded meagre financing for small and privately-owned businesses as one of the major reasons for slowdown in growth. This is where technology can play a major role. With the rise of fintech companies that aim to use technology to improve financial tasks, customers have access to banking services via phone without the need for physical infrastructure as in traditional methods.



(Credit: www.everestgrp.com)

For instance, Alibaba's financial affiliate, Ant Financial Services Group (earlier Alipay), has been one of the pioneers in using artificial intelligence (AI) and Big Data in this sector. Its subsidiary, MyBank, launched four years ago in China, is a cloud-based online banking unit that uses real-time transaction data after permission from borrowers and a risk-management system that analyses more than 3000 parameters before extending loans.